



University of
East London

CSE281 - Introduction to Artificial Intelligence

Car Price Prediction Using Machine Learning

PRESENTED TO: ENG. HEBA GAMAL SALEH

PRESENTED BY: FERASS AHMED MOSTAFA 23P0304

HISHAM MOHAMED ABDRAELRASOUL 23P0259

AHMED MOHAMED ELMLAHE 23P0035

MOHAMED BASSAM 17T0338

Table of Contents

1. Executive Summary	1
2. Introduction	1
3. Dataset Description.....	1
4. Data Preprocessing & Engineering.....	1
5. Exploratory Data Analysis (EDA).....	2
6. Modeling Methodology	2
6.1. Linear Regression (Baseline)	2
6.2. Ridge Regression (L2 Regularization).....	2
7. Hyperparameter Tuning.....	2
8. Model Evaluation	2
9. Conclusion & Future Work.....	3
 Table 1: Results indicate that Ridge Regression outperformed Linear Regression.	2

1. Executive Summary

This report details the design, implementation, and evaluation of a machine learning system capable of predicting used car market prices. By leveraging a dataset of vehicle attributes, we implemented and compared Linear Regression and Ridge Regression models. The study demonstrates that regularization techniques (Ridge) significantly improve prediction accuracy by mitigating multicollinearity, establishing a reliable framework for automated price estimation.

2. Introduction

In the used automotive market, accurate pricing is a critical challenge for both buyers and sellers due to the subjective nature of vehicle valuation. This project aims to solve this by developing a data-driven model that predicts car prices based on objective features such as mileage, model year, and engine volume. Using Python's scikit-learn library, we established a supervised learning pipeline to minimize prediction error and maximize generalization on unseen data.

3. Dataset Description

The analysis utilizes a structured dataset of used car listings containing both numerical and categorical features.

- Target Variable: Price (Continuous).
- Key Predictors:

Categorical: Manufacturer, Model, Fuel Type, Gearbox Type, Drive Wheels, Color.

Numerical: Production Year, Engine Volume, Mileage (Odometer), Cylinders, Airbags.

- Data Split: The data was partitioned into distinct training and testing sets to evaluate model performance objectively.

4. Data Preprocessing & Engineering

Data quality is paramount for regression performance. The following preprocessing pipeline was applied:

- **Data Cleaning:** Irrelevant features and duplicate records were removed to reduce noise.
- **Imputation:** Missing values in critical columns were managed using statistical imputation rather than dropping data.
- **Outlier Removal:** Extreme outliers in Price and Mileage were detected and removed to prevent skewing the linear models.
- **Feature Encoding:** Categorical variables (e.g., Fuel Type, Gearbox) were transformed using One-Hot Encoding to convert text labels into machine-readable numerical vectors.

5. Exploratory Data Analysis (EDA)

Comprehensive EDA was conducted to uncover latent patterns:

- **Correlation Analysis:** A heatmap revealed that Production Year and Engine Volume have a strong positive correlation with Price, while Mileage shows a strong negative correlation.
- **Categorical Insights:** Box plots indicated distinct price tiers based on Transmission type, with automatic vehicles commanding higher market values.
- **Feature Interaction:** We observed that luxury manufacturers maintain higher residual values even with higher mileage.

6. Modeling Methodology

Two supervised regression algorithms were implemented.

6.1. Linear Regression (Baseline)

A standard Ordinary Least Squares (OLS) model was trained to establish a baseline. While interpretable, this model is prone to overfitting when features are highly correlated.

6.2. Ridge Regression (L2 Regularization)

To address the limitations of the baseline, Ridge Regression was implemented. This technique adds a proportional penalty term to the square of the magnitude of coefficients (L2 penalty). This shrinks the coefficients of less key features, reducing model variance.

7. Hyperparameter Tuning

The performance of the Ridge model relies heavily on the regularization strength, alpha.

- **Method:** We performed a grid search over a range of alpha values.
- **Observation:** Low alpha values yielded results like Linear Regression, while extremely high alpha values led to underfitting.
- **Result:** An optimal alpha was selected that minimized the validation error.

8. Model Evaluation

The models were evaluated on the held-out test set using standard regression metrics:

Table 1: Results indicate that Ridge Regression outperformed Linear Regression.

Metric	Description	Performance (Ridge vs Linear)
MAE	Mean Absolute Error	Lower in Ridge (Better)
MSE	Mean Squared Error	Lower in Ridge (Better)
R ² Score	Variance explained	Higher in Ridge (Better fit)

9. Conclusion & Future Work

The project successfully deployed a machine learning model capable of estimating car prices with high reliability. The transition from Linear to Ridge Regression provided a measurable improvement in generalization.

Future enhancements include:

- Ensemble Methods (Random Forest/XGBoost).
- Deployment via Flask API.
- Advanced Feature Engineering.